

SET UP DATABASE SCHEMA

Before we begin code, let's first create our database schema. Here we presume a simple users table having username, email and password.

Script below should help do that.

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) NOT NULL UNIQUE,  
  password VARCHAR(255) NOT NULL,  
  role ENUM('admin', 'user') DEFAULT 'user',  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP  
);
```

```
INSERT INTO users (name, email, password, role) VALUES  
(  
  'Christelle Sikuli', 'christelle.sikuli@example.com',  
  '$2b$10$abc1234567890examplehashvalue1234567890', 'admin'),  
(  
  'Mugisha Amos', 'mugisha.amos@example.com', '$2b$10$abc1234567890examplehashvalue1234567890',  
  'user'),  
(  
  'Opolot Jonathan', 'opolot.jonathan@example.com',  
  '$2b$10$abc1234567890examplehashvalue1234567890', 'user'),  
(  
  'Isaac Kayemba', 'isaac.kayemba@example.com', '$2b$10$abc1234567890examplehashvalue1234567890',  
  'user');
```

CREATE OUR ENVIRONMENTAL VARIABLES

Lets setup our environmental Variable, but what is it really?

These are configuration values that your app needs, but you **don't want to hardcode directly into your source code.**

In JS, we use a .env configuration file to store that. Below is an example of the configuration file.

```
DB_HOST=localhost
```

```
DB_PORT=3306
```

```
DB_USER=root
```

```
DB_PASSWORD=mysecret
```

```
DB_NAME=school_db
```

CREATE AND ESTABLISH CONNECTION TO THE DATABASE

Now that our environment is set, lets connect to our MySQL database.

Create a JS file, give a name of choice, but remember names should aswell make sesnse. I would suggest something like db.js, database.js, mysql_connection.js, etc.

See below how to connect.

```
const mysql = require("mysql2/promise");
```

```
const dotenv = require("dotenv");
```

```
dotenv.config();
```

```
const pool = mysql.createPool({
```

```
  host: process.env.DB_HOST,
```

```
  port: process.env.DB_PORT,
```

```
  user: process.env.DB_USER,
```

```
  password: process.env.DB_PASSWORD,
```

```
  database: process.env.DB_NAME,
```

```
});
```

```
module.exports = pool;
```

Feel free to run your file in the terminal to see if it throws any errors.

CREATE YOUR FIRST ROUTE

Now, that you don't have any errors,

Lets create our first API that fetches all users from the database.

```
const pool = require("./db.js");
const express = require("express");

const app = express();

app.use(express.json());

// GET ALL USERS
app.get("/users", async (req, res) => {
  try {
    const usersQuery = "SELECT * FROM users";
    const [rows] = await pool.query(usersQuery);
    res.json(rows);
  } catch (err) {
    console.error("Error fetching users....", err);
    res.status(500).json({ error: "Failed to Fetch Users" });
  }
});

const PORT = 3000;

app.listen(PORT, () => console.log(`Server running on ${PORT}`));
```

Now run `node app.js`, any errors? Debug 😊 it

YOU ARE DONE!!

So, lets move to post man and test our API for getting all users from the database using this link.

localhost:3000/users

the link above also can return data to the we browser, you can try.